

Sweta Pati

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(Open to Relocate)

SUMMARY

AI/ML Engineer with 5+ years of experience building production **ML, NLP, and GenAI systems** for Fortune 500 clients across **finance, insurance, automation, and information extraction**. Expertise in **multi-agent systems, LLM workflows, RAG, model fine-tuning, evaluation, observability, and MLOps**. Published **event extraction research in ACL** and a track record of delivering measurable business impact.

SKILLS

- **Programming Languages:** Python, SQL, PyTorch, Scikit-learn, Pandas, NumPy
- **Agentic and RAG:** LangChain, LangGraph, Deep Agents, Evaluation, Observability, Harness Engineering
- **NLP & GenAI:** Transformers, Fine-tuning, PEFT (LoRA/QLoRA), RLHF, Prompt Engineering
- **Machine Learning & Deep Learning:** Supervised, Classification, Regression, Ensemble Model, Neural Networks
- **Data Analysis and Visualization:** Exploratory Data Analysis (EDA), Tableau, Matplotlib
- **MLOps & Deployment:** Docker, FastAPI, Azure,, GitHub, MLflow, Weights & Biases

PROFESSIONAL EXPERIENCE

CENTRAL MUTUAL INSURANCE

Ohio, USA

Machine Learning Engineer

Jan 2026 - Present

Architecting and building production AI solutions to automate insurance workflows and improve operational efficiency.

Project 1: Rate Manual Comparison: Automating insurance rate manual comparison to identify meaningful changes in rules, rates, and policy structure.

- **Built a LLM-powered document comparison pipeline for 50–300+ page insurance manuals**, reducing manual comparison from **5 days to 30minutes** using **Azure Document Intelligence, GPT-4o, and Python**.
- Engineered a **hybrid LLM and deterministic workflow** combining document parsing, regex section boundary detection, python difflib comparison with LLM summary to identify meaningful policy differences.
- **Productionized Streamlit** application on **Kubernetes** with **Azure Blob Storage**, and **Azure DevOps CI/CD** across **19 lines of business**, preserving rule hierarchy, section boundaries.

Project 2: Claims Leakage Agent: Developing a multi-agent system to identify potential claims leakage through evidence-grounded auditing and human-in-the-loop review.

- **Shipped a production multi-agent Claims Leakage platform** with **LangChain Deep Agents** on **LangGraph**, orchestrating **8 domain-specialist agents concurrently** and cutting manual claim-review effort by **90%**.
- Engineered the agent harness with **timeouts, retries, Pydantic-validated outputs, deterministic code-owned logic, partial-run recovery, and 18 live MCP tools** for reliable long-running execution.
- **Implemented end-to-end agent observability** with **OpenTelemetry, Azure Application Insights, and Microsoft Foundry**, tracing agent, tool execution, latency, failures, token usage, and cost.
- **Built an agent evaluation framework** across **100 ground-truth claims** using deterministic checks and human-calibrated **LLM-as-judge** scoring to measure finding quality and consistency.
- **Optimized context engineering** for **200+ page policies** using **form preclassification actor-verifier agents, tool-result offloading, and prompt caching**, reducing context pressure and execution latency.
- **Improved factual reliability** and reduced **hallucinations** through **source-grounded attribution**, domain-aware **actor-verifier agents**, and **human-in-the-loop review**, requiring findings to trace back to claim data.
- Productionized the **Streamlit application on on-prem Kubernetes** using **Azure DevOps CI/CD**.

4A CONSULTING

AI/ML Engineer Associate - Intern

Maryland, US

June 2025 - Jan 2026

Driving efficiency by building AI-driven recruitment automation systems, integrating multi-agent orchestration and RAG.

- **Improved interview scoring consistency by 30%** and mitigated bias across **100+ assessments** by productionizing an **LLM-based evaluation pipeline** using **OpenAI APIs, Python, and rubric-guided scoring**.
- Built a **RAG system** using **LangChain, FAISS, and OpenAI embeddings** to reduce manual lookup time for FAQs and documentation across **4000+ queries a month**, evaluated with **200 gold test samples**.
- Deployed scorer modules by **containerizing** with **Docker**, exposing via **FastAPI**, and hosting on **Azure**.
- Developed a **multi-agent framework (LangGraph, MCP)**, orchestrating **documentation query retrieval, status checker, score interpreter** agents, automating recruiter workflows and enhancing efficiency.

- Increased **cost attribution** by **25%** by implementing **token-level tracking** with feature tagging.
- **Aligned AI feature** integration with **product roadmaps**, collaborating with **cross-functional teams**.

GEORGE MASON UNIVERSITY

Natural Language Processing Graduate Research Assistant

Virginia, US

May 2024 - Jan 2025

Advance state-of-the-art event extraction research by fine-tuning LLMs with schema-guided instruction tuning.

- Co-authored **ACL 2025** paper Instruction-Tuning LLMs for **Event Extraction** with Annotation Guidelines, advancing schema-guided **instruction-tuning** with **Python code prompts** and annotation guidelines.
- Achieved **+10% Trigger** and **+5% Argument** Classification **F1 gains** on **ACE05** and **RichERE** benchmarks by combining annotation guidelines with contrastive training.
- Fine-tuned **LLMs** (LLaMA-3.1-8B, LLaMA-3.2-1B, Qwen2.5-Coder-1.5B) using **LoRA** with 4-bit quantization, **PEFT**, **PyTorch**, Unsloth, and **Hugging Face**, improving event extraction performance.
- Built a custom **evaluation framework** with schema validation, assessing model outputs against gold labels.
- Enhanced **cross-domain generalization**, replacing expert-written rules with **GPT-4o-generated** annotation **guidelines** for **500+ events** and **4,000+ arguments**, tracked **100+ training runs** in **Weights & Biases**.

HIGHRADIUS TECHNOLOGIES PRIVATE LIMITED (FinTech)

Software Engineer – II (Data Science)

Odisha, India

Jul 2019 - Aug 2023

Clients: Kraft Heinz, Clorox, L'Oréal, Ferrero, Duracell

Led a 10-member team delivering production AI/ML solutions for 15+ Fortune 500 clients and automating ML pipelines to accelerate revenue recovery, reduce payment delays, and prevent order disruptions.

- **Increased dollar recovery rate** by **12%** with **85% recall** across **15+ Fortune 500** clients by developing **classification models** (LGBM, Random Forest) for **fraud detection** with **SHAP** interpretability.
- **Reduced manual effort** by **25%** by automating **AI action code assignment** and **worklist prioritization**.
- **Increased on-time collections** by **17%** by predicting invoice payment dates with **regression models** (XGBoost, Random Forest) using payment history and invoice metadata, evaluated via **RMSE**.
- **Improved collections efficiency** by **20%** by deploying **AI-prioritized collector worklists** and **suggested actions**, enabling personalized collection strategies and visualizing outcomes in **Tableau dashboards**.
- **Reduced model implementation** and **deployment time** by **40%** by building **automated ML pipelines** for EDA, feature engineering, training, and hyperparameter tuning, deployed via **AWS (S3, SageMaker)**.
- Enhanced experiment tracking with **MLflow**, comparing **100+ model runs** and their hyperparameters.

PUBLICATIONS, CERTIFICATIONS, AWARDS

- Published paper: [Instruction-Tuning LLMs for Event Extraction with Annotation Guidelines](#) at **ACL 2025**
- AI Agents Fundamentals, Hugging Face [[Credential](#)]
- Google AI Essentials, Google [[Credential](#)]
- Finetuning Large Language Models [[Credential](#)]
- Awarded as Best Performer of the Team, associated with **Highradius Technologies**

PROJECTS

FinInsight – Earnings Insight RAG Assistant [[Source Code](#)]

Tech Stack: Python, LangChain, OpenAI (text-embedding-3-small), GPT-3.5, Pinecone, PyMuPDF, RecursiveTextSplitter

- Built a **RAG pipeline** to parse, embed, and query 10-K/10-Q earnings reports with **LangChain**, **OpenAI embeddings**, and **Pinecone**, enabling traceable source-linked natural language Q&A over financial filings.

Instruction-Tuning LLMs for Event Extraction (ACL 2025 Findings) [[Paper](#) | [Source Code](#) | [Slides](#)]

Tech Stack: Python, PyTorch, Hugging Face, PEFT, LLaMA-3.1, Qwen2.5-Coder, GPT-4o, Weights & Biases

- **Fine-tuned** LLaMA-3 and Qwen2.5 with schema-guided instruction tuning and GPT-4o annotation guidelines.
- Achieved **+10% Trigger** and **+5% Argument F1 gains** on **ACE05** and **RichERE** benchmarks, **improving cross-domain** and **low-resource generalization**.

EDUCATION

GEORGE MASON UNIVERSITY

Master's in Computer Science (GPA: 3.93/4)

Virginia, US

Aug 2023- May 2025

KIIT UNIVERSITY

Bachelor's in Computer Science

Odisha, India

May 2016 - May 2020